CoreLog Vision: Lithology from Drill-Core Imagery, with Honest Sim-to-Real Out-of-Distribution Detection

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Abstract—**Logging the lithology of drill core, identifying the rock type down a borehole from the physical core in its trays, is slow, subjective manual work, and automating it from core-tray photographs is an obvious target for computer vision. The catch is data: labelled core photographs are scarce, so a model is often trained on synthetic or one operation's core and then asked to read another's, and it will do so *silently and wrongly* unless it can tell that the new core is unlike its training data. CoreLog Vision is an in-browser pipeline that classifies core-tray imagery into lithologies and, crucially, detects when it is looking at core it was not trained on. On synthetic core its convolutional classifier reaches 99.4% accuracy over six lithologies against a classical colour-and-texture baseline's 92.9%, on a leakage-safe split that holds out entire boreholes (not random patches) so the number is not inflated by patches from the same hole appearing in train and test. On real core, from the open DCID-7 drill-core image dataset, a dedicated head reaches 99.2% over seven lithologies, while the synthetic-trained features transfer to only 88.8%, the honest sim-to-real gap. The report's central result is the detector that guards that gap: nine out-of-distribution methods are benchmarked at telling real DCID-7 core from synthetic training core (both funnelled to the same 24-pixel window so no detector can cheat on resolution), and *feature-space* detectors succeed (the best at area-under-precision-recall 0.999, the shipped one at 0.92) while a *reconstruction*-based detector fails (0.28). The lesson, know what you do not know, and do it in feature space, is the report's contribution; every number and figure regenerates from the committed artifacts.**

Index Terms—**Drill-core logging, lithology classification, computer vision, convolutional neural networks, out-of-distribution detection, sim-to-real, Mahalanobis, mineral exploration, reproducibility.**

I. Introduction

A Geologist logs a drill hole by reading its core, tray by tray, and naming the rock type at each depth, a task that is slow, tiring and subjective, and that produces the lithology column every downstream mine-planning decision rests on. Photographs of the core trays are routinely taken, so classifying lithology from those images with computer vision is a natural automation, and convolutional networks do it well [1], [2], [4], [5]. The obstacle is not the classifier but the data. Labelled core imagery is scarce and operation-specific: core from one deposit, one drill, one photographic setup looks different from another's, so a model trained on synthetic core or on one mine's photographs faces a distribution shift when applied elsewhere. A classifier will answer confidently on such out-of-distribution core and be silently wrong, which in a logging context means a corrupted lithology column no one flagged.

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Technical report, 2026. Interactive tool and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_CoreLog; live at https://corelog.fasl-work.com.

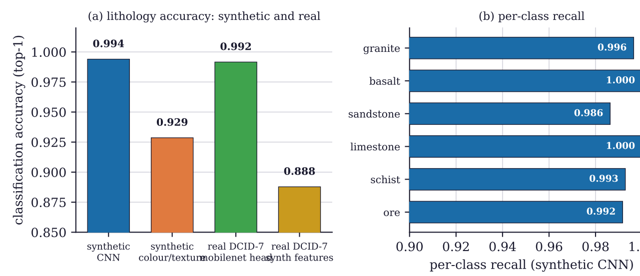

Fig. 1. Lithology classification accuracy. **(a)** On synthetic core the convolutional network reaches 99.4% against a classical colour-and-texture baseline's 92.9%, on a leakage-safe split that holds out entire boreholes; on real DCID-7 core a dedicated head reaches 99.2% over seven lithologies, while the synthetic-trained features transfer to only 88.8%, the honest sim-to-real gap. **(b)** The synthetic CNN's per-class recall is 0.986 to 1.000 across the six lithologies, with no weak class.

CoreLog Vision is a pipeline built around that problem. It segments a core-tray image along each core channel, classifies each window's lithology with a small convolutional network run live in the browser, and stitches a depth log, and it does two honest things the classifier alone does not. First, it is evaluated without leakage, on a split that holds out entire boreholes, so the accuracy is what the model would achieve on a genuinely new hole, not an inflated number from patches of the same hole on both sides of the split. Second, and centrally, it carries an out-of-distribution detector that flags when the incoming core is unlike the training data, so a user is warned rather than misled. This report reports the classifier's honest accuracy on synthetic and on real core, and benchmarks the out-of-distribution detector, with the finding that feature-space detection works where reconstruction-based detection does not.

II. The Lithology Classifier

The classifier is a small convolutional network over a sliding window along each core channel, classifying into six

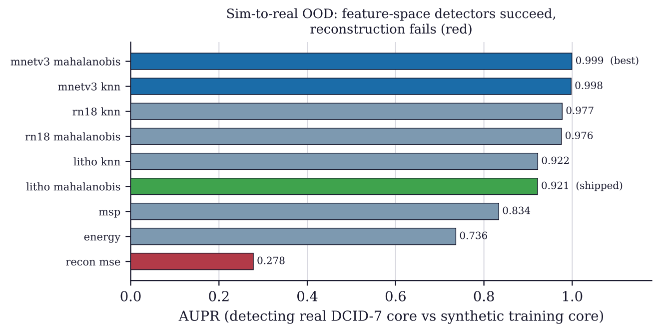

Fig. 2. Sim-to-real out-of-distribution detection: nine detectors telling real DCID-7 core from the synthetic training core, scored by area under the precision-recall curve. Both streams are funnelled to the same 24-pixel window so no detector can exploit a raw resolution or blur difference. *Feature-space* detectors (Mahalanobis and k-nearest-neighbour distances in a network's feature space) succeed, the best at 0.999 and the shipped lightweight one at 0.92; softmax-based scores are moderate; and a *reconstruction*-error detector fails outright (0.28).

lithologies (granite, basalt, sandstone, limestone, schist, ore) with a softmax confidence, trained offline and run live in the browser. Its accuracy is reported on a split that holds out entire boreholes, fourteen holes to train and four to test, so no window from a test hole is ever seen in training; this matters because random patch splits inflate core-logging accuracy by letting near-identical patches of one hole appear on both sides. On that honest split (Fig. 1) the network reaches 99.4% accuracy, against 92.9% for a classical colour-and-texture baseline, a 6.5-point margin that is the value of learning texture over hand-crafted colour statistics, and its per-class recall is uniformly high (0.986 to 1.000), with no lithology the model systematically misses. On synthetic core, in short, the classifier is excellent and honestly so.

III. Real Core and the Sim-to-Real Gap

Synthetic core is not the target; real core is, and here the honesty of the report matters. Applied to the open DCID-7 drill-core image dataset (seven real lithologies, from red sandstone through granite and basalt to marble), the pipeline shows the expected sim-to-real behaviour (Fig. 1a). A *dedicated* head, a compact MobileNetV3 [3] trained on the real data, reaches 99.2% top-1 accuracy over the seven lithologies, genuinely strong on real core. But the features learned on *synthetic* core, transferred to the real data with a light head, reach only 88.8%: the ten-point drop is the sim-to-real gap, the price of having been trained on procedural core rather than photographs. Both numbers are honest and both are useful: they say that real core is learnable to high accuracy *if* you have real labels, and that a synthetic-trained model degrades meaningfully on real core, which is exactly why the pipeline must know when it is seeing real core.

IV. Detecting Out-of-Distribution Core

The guard against silent sim-to-real error is an out-of-distribution detector, and its design is the report's central methodological result (Fig. 2). Nine detectors are benchmarked on the task of distinguishing real DCID-7 core (out-of-distribution) from held-out synthetic core (in-distribution), with a control that removes the obvious cheat: both streams are funnelled through the same 24-pixel live-window resolution, so no detector can win merely by noticing that real photographs are sharper or blurrier than synthetic ones. Under that fair comparison, the ranking is clear and instructive. *Feature-space* distance detectors, a Mahalanobis or k-nearest-neighbour distance in the feature space of a network [7], dominate: the best (a MobileNetV3 feature Mahalanobis) reaches area-under-precision-recall 0.999, and even the lightweight shipped detector, a Mahalanobis distance in the lithology CNN's own 64-dimensional features, reaches 0.92 (AUROC 0.95). Softmax-confidence detectors [6] and the energy score [8] are moderate (0.74 to 0.83). And a *reconstruction*-error detector, an autoencoder flagging what it cannot reconstruct, fails almost completely (0.28), below the useful range. The lesson is specific and portable: to know that incoming core is unlike the training data, measure distance in a good feature space, not reconstruction error, and a lightweight feature-Mahalanobis detector reusing the classifier's own features is enough to ship.

V. Discussion and Scope

The pipeline's three numbers tell a coherent, honest story. On synthetic core the classifier is excellent (99.4%) and beats the classical baseline; on real core it is excellent *with* a dedicated head (99.2%) and degrades to 88.8% on transferred synthetic features; and it can tell real core from synthetic at area-under-precision-recall up to 0.999 in feature space. Together they describe a system that is accurate where it has been trained and, crucially, aware of when it has not, which is the property a deployable core-logging aid needs. The scope is stated plainly. The six-class synthetic accuracy is on procedurally generated core, useful for controlled evaluation but not a photograph of real rock; the real numbers are on the open DCID-7 dataset, a genuine but single dataset, and a production deployment would need calibration and labels from the target operation. The tool is a logging *aid* with an out-of-distribution guard, not an autonomous logger, and it says so, and it flags the core it is unsure about rather than logging it confidently. None of this is hidden; the interface shows the confidence and the out-of-distribution flag on every segment.

VI. Conclusion

CoreLog Vision classifies drill-core lithology from tray imagery, at 99.4% on synthetic core (against a classical baseline's 92.9%, leakage-safe) and 99.2% on real DCID-7 core with a dedicated head, and it guards the sim-to-real gap with an out-of-distribution detector that reaches 0.999 in feature space where a reconstruction detector fails. Reporting the leakage-safe accuracy, the honest sim-to-real drop, and the feature-space out-of-distribution result together, so the system knows what it does not know, is the report's discipline, and the reproducible pipeline that does it is its contribution.

Appendix A
Reproduction

The committed artifacts under `data/derived/` carry the lithology-classifier metrics (`cl-learned.json`: the

CNN and baseline accuracy on the borehole-grouped split, the per-class confusion) and the out-of-distribution benchmark (`ood-bench.json`: the nine detectors, the resolution control, the shipped and best detectors, and the real-head accuracies on DCID-7). The figure scripts under `manuscripts/corelog-vision/figures/` read those artifacts. The models are trained offline (PyTorch, exported to ONNX) and run in the browser (onnxruntime-web); re-evaluating reproduces Figs. 1–2.

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